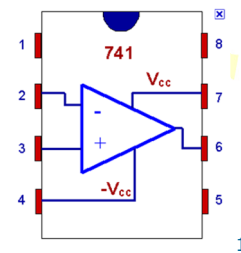
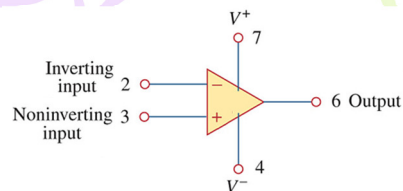


EBN 122 ELECTRICITY AND ELECTRONICS



Chapter 5 Operational Amplifiers Op Amps



Operational Amplifier - Chapter 5

5.2 What is an Op Amp?

5.3 Ideal Op Amp

5.4 Configuration of Op Amp

5.5 Cascaded Op Amp

5.6 Application

– Digital-to Analog Converter

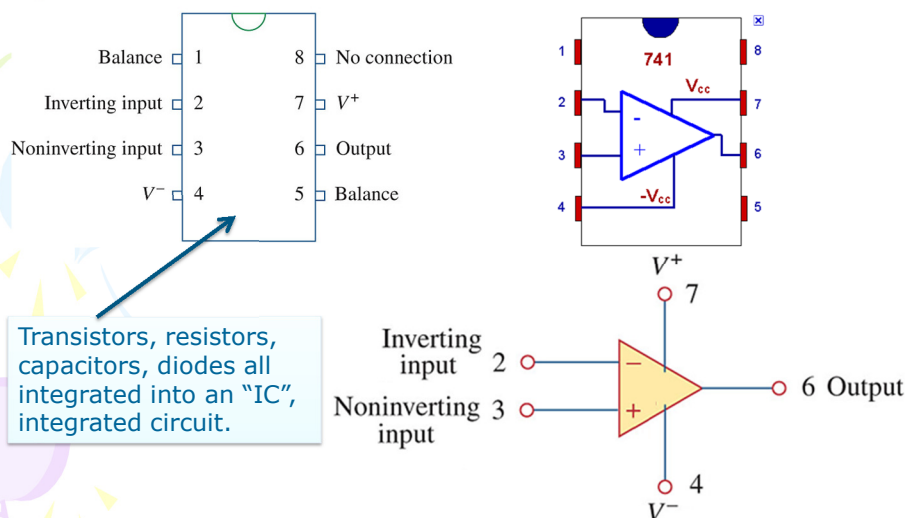
5.2 What is an Op Amp

- It is an electronic unit that behaves like a voltage-controlled voltage source.
- It is an active circuit element designed to perform mathematical **operations** of **addition, subtraction, multiplication, division, differentiation** and **integration**.
- Hence: **OPERATIONAL** Amplifier

3

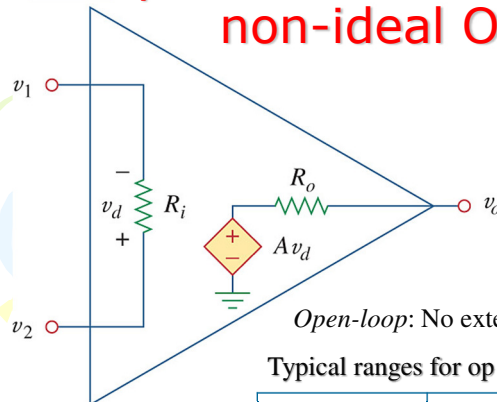
5.2 What is an Op Amp

Pin Configuration and Circuit Symbol



5.2 What is an Op Amp

Equivalent Circuit model of a non-ideal Op Amp



Op Amp output:
 v_o as a function of v_d

$$v_d = v_2 - v_1$$

$$v_o = Av_d = A(v_2 - v_1)$$

$A = \text{Open-loop voltage gain}$

Open-loop: No external feedback from output v_o to input v_d

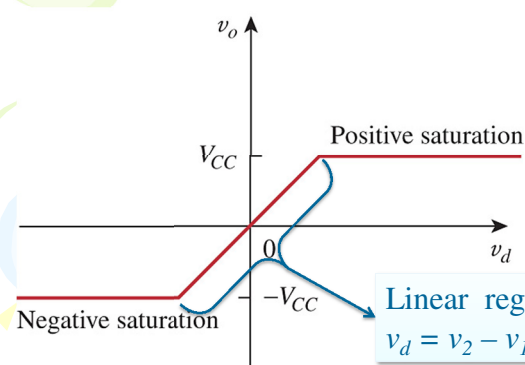
Typical ranges for op amp parameters

Parameter	Range	Ideal
A	$10^5 - 10^8$	∞
R_i	$10^5 - 10^{13} \Omega$	$\infty \Omega$
R_o	$10 - 100 \Omega$	0Ω
Supply voltage $\rightarrow V_{CC}$	$5 - 24 \text{ V}$	

5

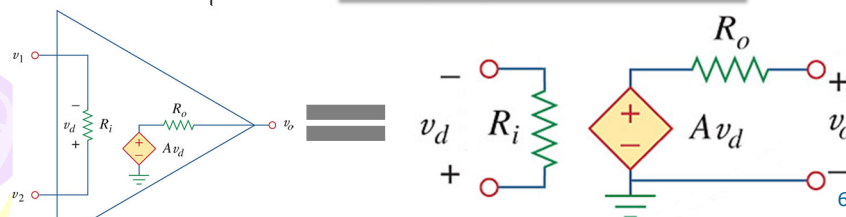
5.2 What is an Op Amp

A Practical Limitation



Output cannot exceed $|V_{CC}|$

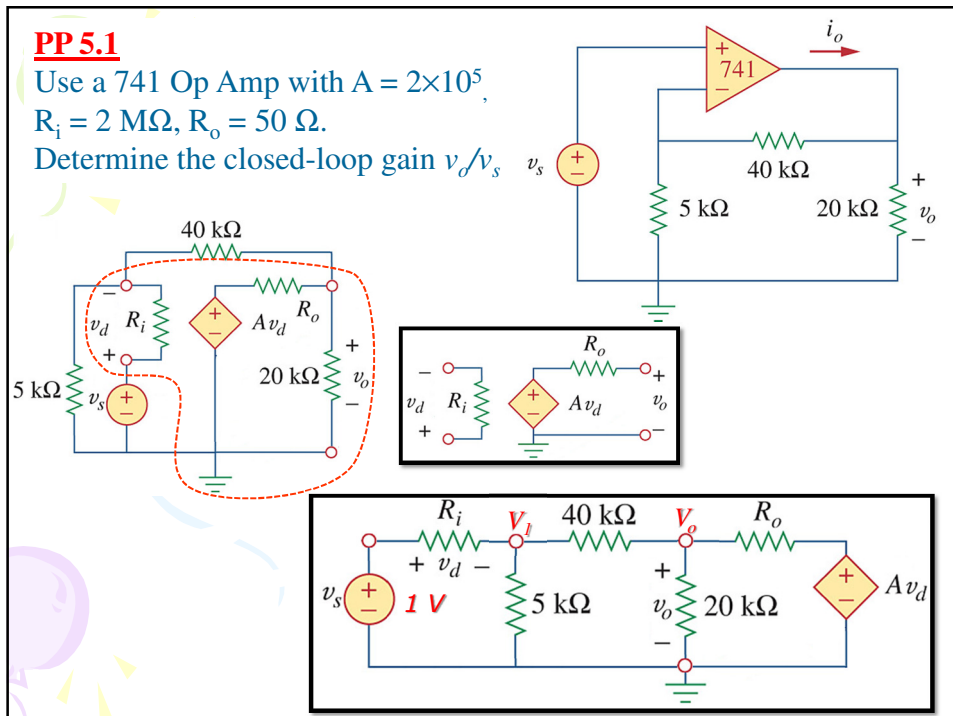
Linear region where $v_o = Av_d$
 $v_d = v_2 - v_1 = v^+ - v^-$



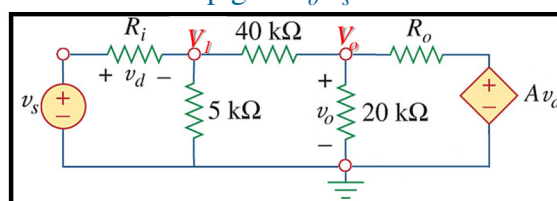
PP 5.1

Use a 741 Op Amp with $A = 2 \times 10^5$,
 $R_i = 2 \text{ M}\Omega$, $R_o = 50 \text{ }\Omega$.

Determine the closed-loop gain v_o/v_s .

**PP 5.1 cont...**

Use a 741 Op Amp with $A = 2 \times 10^5$, $R_i = 2 \text{ M}\Omega$, $R_o = 50 \text{ }\Omega$.
 Determine the closed-loop gain v_o/v_s .



Nodal analysis:

KCL @ Node V_1

$$\frac{V_1 - V_s}{2 \times 10^6} + \frac{V_1}{5 \times 10^3} + \frac{V_1 - V_o}{40 \times 10^3} = 0$$

$$V_1 - V_s + 400V_1 + 50V_1 - 50V_o = 0$$

$$451V_1 - 50V_o = V_s \quad (1)$$

KCL @ Node V_o

$$\frac{V_o - V_1}{40 \times 10^3} + \frac{V_o}{20 \times 10^3} + \frac{V_o - Av_d}{50} = 0$$

$$V_o - V_1 + 2V_o + 800[V_o - A(V_s - V_1)] = 0$$

$$160 \times 10^6 V_1 + 803V_o = 160 \times 10^6 V_s$$

$$451V_1 + 2.263 \times 10^{-3}V_o = 451V_s \quad (2)$$

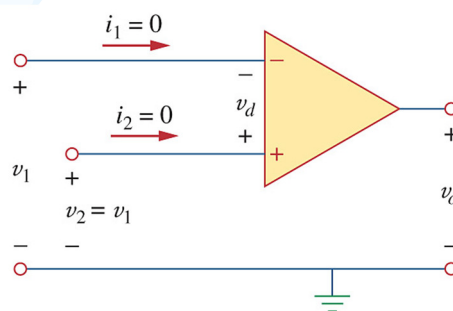
(1) - (2)

$$\begin{aligned} -50.00226 &= -450V_s \\ \frac{v_o}{v_s} &= \frac{-450}{-50.00226} = 8.9996 \end{aligned}$$

5.3 Ideal Op Amp

An ideal op amp has the following characteristics:

1. Infinite open-loop gain, $A \approx \infty$
2. Infinite input resistance, $R_i \approx \infty$
3. Zero output resistance, $R_o \approx 0$



Since $R_i = \infty$:

$$i_1 = i_2 = 0$$

$$\therefore v_d = v_2 - v_1 = 0$$

$$\therefore v_2 = v_1$$

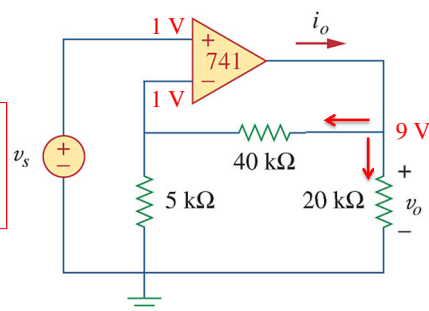
9

PP 5.1 for ideal opamp

Assume $v_s = 1\text{ V}$ and determine v_o and i_o .

$$\begin{aligned} v_+ &= 1\text{ V} = v_- \\ i_1 &= i_2 = 0 \\ \therefore v_d &= v_2 - v_1 = 0 \\ \therefore v_2 &= v_1 \end{aligned}$$

Remember:
 $i_1 = i_2 = 0$
 $v_+ = v_-$



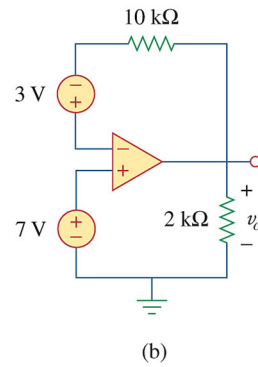
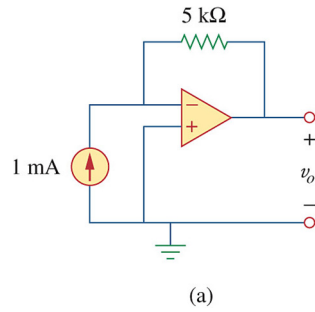
KCL @ Node v_- to get V_o :

$$\begin{aligned} \frac{1}{5 \times 10^3} + \frac{1 - V_o}{40 \times 10^3} &= 0 \\ 8 + 1 - V_o &= 0 \\ V_o &= 9\text{ V} \end{aligned}$$

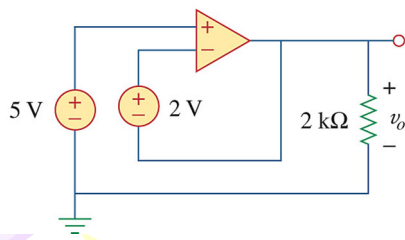
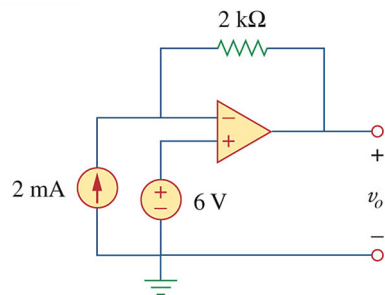
KCL @ Node V_o to get i_o :

$$\begin{aligned} i_o &= \frac{9 - 1}{40 \times 10^3} + \frac{9}{20 \times 10^3} \\ i_o &= 650\text{ }\mu\text{A} \end{aligned}$$

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Problem 5.8Determine v_o for both circuits

11

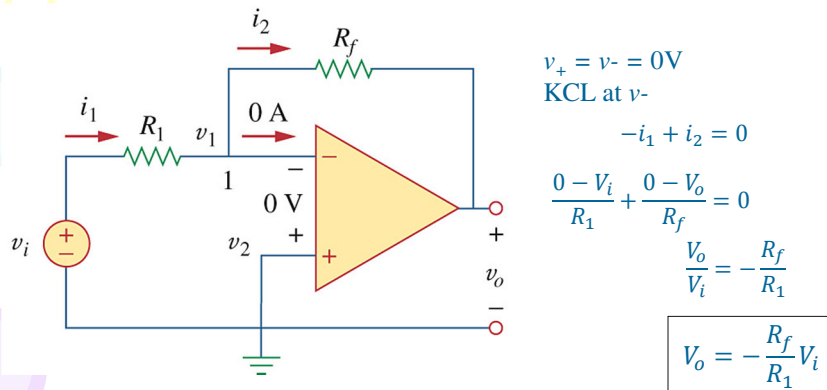
Problem 5.9Determine v_o for both circuits

12

Ch. 5 L#2

5.4 Inverting Amplifier

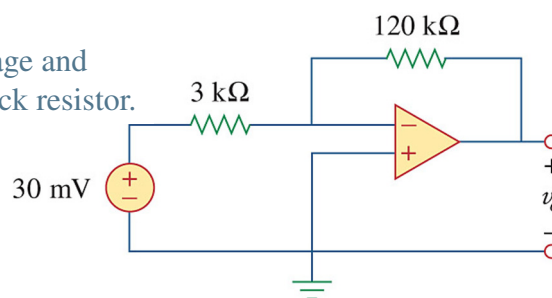
- Inverting amplifier reverses the polarity of the input signal while amplifying it



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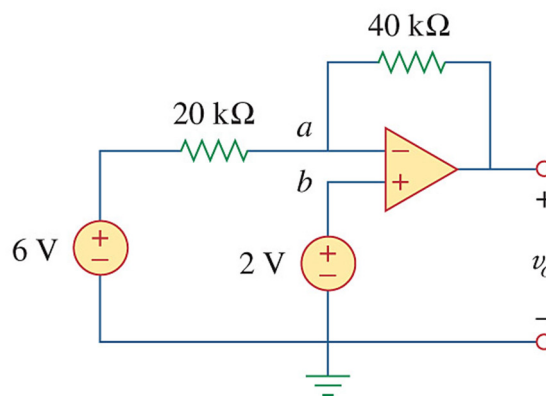
PP 5.3

Determine the output voltage and current through the feedback resistor.



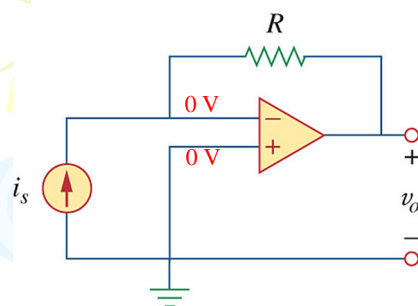
14

Example 5.4
Determine V_o .



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5.4 Inverting Amplifier
Special Case: Current-to-Voltage converter



$$v_+ = v_- = 0V$$

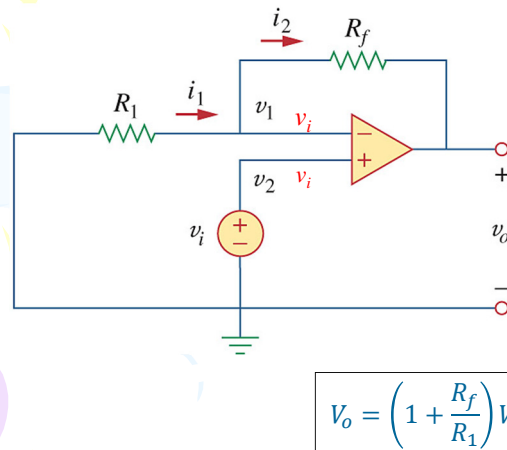
$$\text{KCL at } v_-: \quad i_s = \frac{0 - v_o}{R}$$

$$V_o = -i_s R$$

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5.5 Non-inverting Amplifier

- Non-inverting amplifier is designed to produce positive voltage gain



$$v_+ = v_- = V_i$$

KCL at v_-

$$\frac{V_i - 0}{R_1} + \frac{V_i - V_o}{R_f} = 0$$

$$\frac{V_i - V_o}{V_i} = -\frac{R_f}{R_1}$$

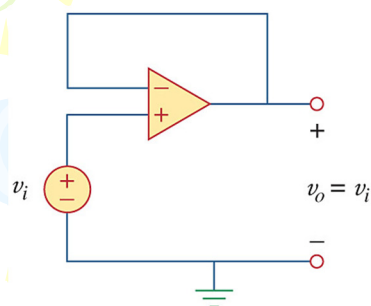
$$1 - \frac{V_o}{V_i} = -\frac{R_f}{R_1}$$

$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}$$

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_i$$

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5.5 Non-inverting Amplifier Special Case: Voltage Follower



- Output = Input
- Advantage: V_i sources a very low current while V_o can source high current. (Remember Op Amp is an Active element getting energy via V_{CC} and $-V_{CC}$).
- Circuit is used as a BUFFER to isolate one stage of a circuit from another.

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PP 5.5Calculate v_o .

$$i_+ = 0 \text{ A}$$

Because of this, the 3V is divided across the 4 k Ω and 8 k Ω resistors, hence a basic voltage divider.

So:

$$v_+ = 3 \frac{8}{4+8} = 2 \text{ V} = v_-$$

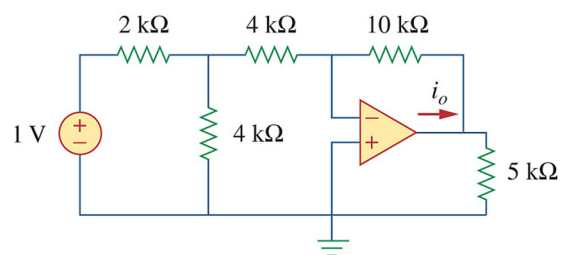
KCL @ v_- :

$$\frac{2-0}{2k} + \frac{2-V_o}{5k} = 0$$

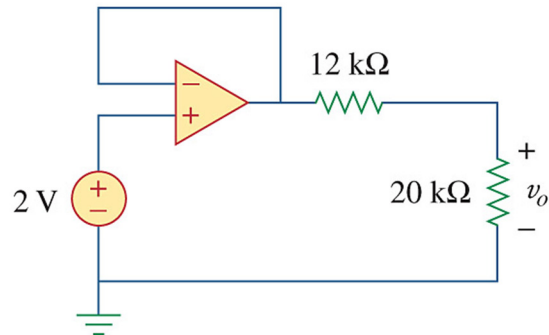
$$10 + 4 - 2V_o = 0$$

$$V_o = 7 \text{ V}$$

19

Problem 5.19Calculate i_o .

20

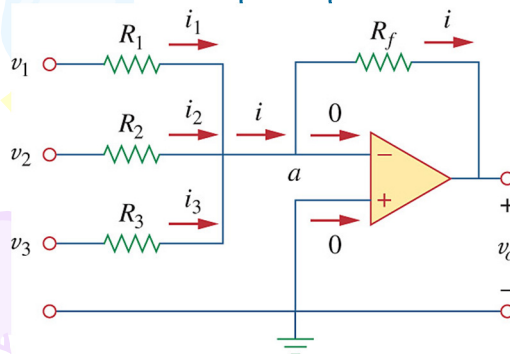
Problem 5.25Calculate v_o .

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Ch. 5 L#3

5.6 Summing Amplifier

- Summing Amplifier is an op amp circuit that combines several inputs and produces an output that is the weighted sum of the inputs.
- Basic configuration of an inverting amplifier but with multiple inputs.



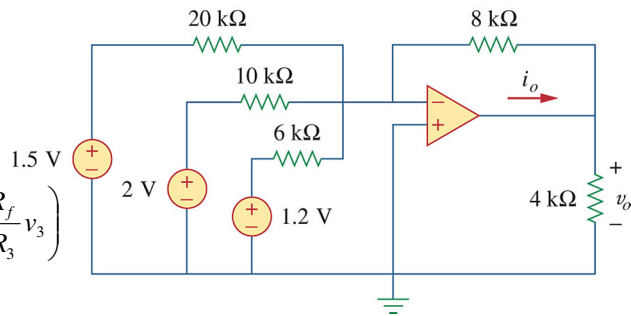
$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$

22

PP 5-6

Determine v_o and i_o for the following circuit:

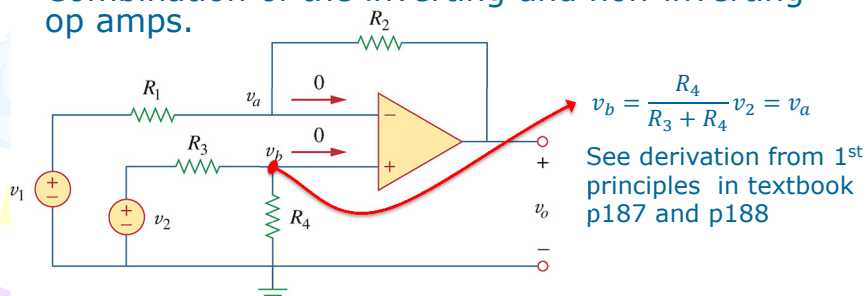
$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$



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Ch. 5 L#3**5.7 Difference Amplifier**

- Difference amplifier is a device that amplifies the difference between two inputs but rejects any signals common to the two inputs.
- Combination of the inverting and non-inverting op amps.



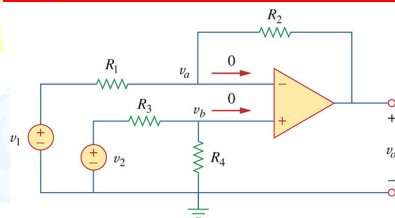
$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)}v_2 - \frac{R_2}{R_1}v_1$$

1. If $R_1/R_2 = R_3/R_4$: $V_o = 0$ if $V_1 = V_2$
2. If $R_2 = R_1$ and $R_3 = R_4$: $V_o = V_2 - V_1$

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Example 5.7

Design an Op Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$

Design 1: Using only 1 Op Amp

$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

$$v_o = 3v_2 - 5v_1$$

$$\frac{R_2}{R_1} = 5$$

$$\frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} = \frac{5(1 + 1/5)}{1(1 + R_3/R_4)} = 3$$

$$\frac{5(1 + 1/5)}{3} = (1 + R_3/R_4)$$

$$2 - 1 = 1 = R_3/R_4 \quad \text{Choose } R_3 = R_4 = 2 \text{ k}\Omega$$

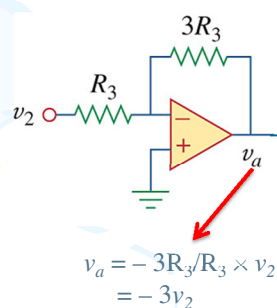
25

Example 5.7

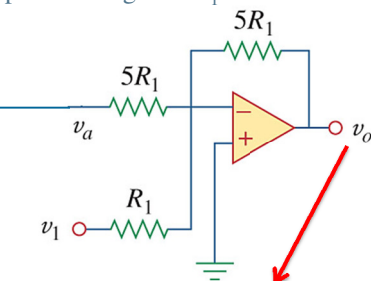
Design an Op Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$

Design 2: Using 2 Op Amps

1. Build an inverting amp to obtain $v_a = -3v_2$.
2. Feed the output v_a into a summing amplifier along with v_1 .



$$v_a = -3R_3/R_3 \times v_2 = -3v_2$$

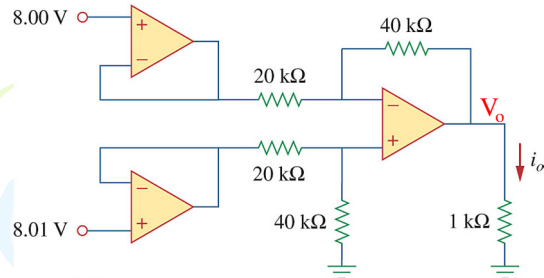


$$\begin{aligned} v_o &= -(5R_1/R_1 \times v_1 + 5R_1/5R_1 \times v_a) \\ &= -5v_1 - v_a \\ &= -5v_1 - (-3v_2) \\ &= -5v_1 + 3v_2 \end{aligned}$$

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PP 5-8

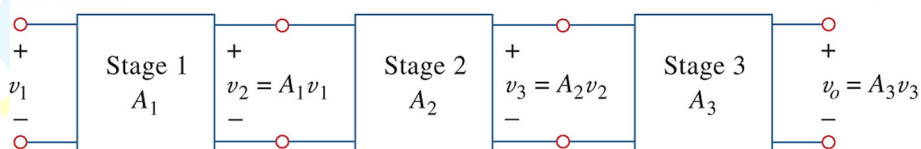
Determine i_o for the following circuit:



27

Ch. 5 L#4**5.8 Cascaded Op Amp Circuits**

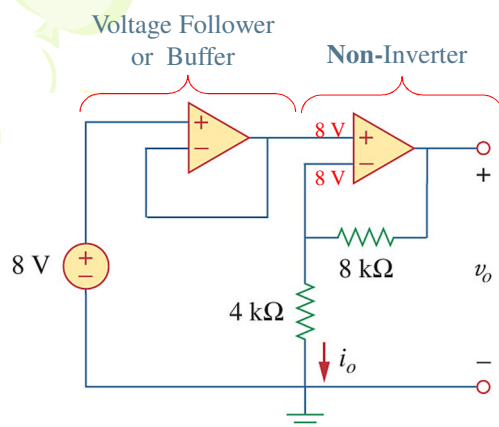
- It is a head-to-tail arrangement of two or more op amp circuits such that the output to one is the input of the next.



$$v_o = A_3 v_3 = A_3 (A_2 v_2) = A_3 A_2 (A_1 v_1)$$

$$\text{Total Amplification } A = A_1 A_2 A_3$$

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PP 5-9Determine v_o and i_o .

$$v_+ = v_- = 8 \text{ V}$$

KCL at v_-

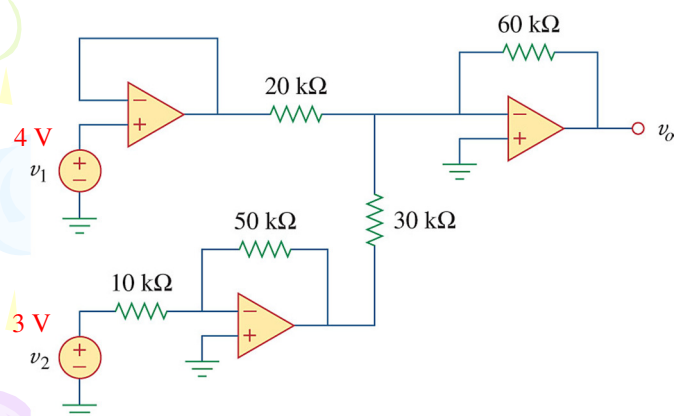
$$\frac{8 - 0}{4k} + \frac{8 - v_o}{8k} = 0$$

$$16 + 8 = v_o$$

$$v_o = 24 \text{ V}$$

$$i_o = \frac{v_-}{4k} = \frac{8}{4k} = 2 \text{ mA}$$

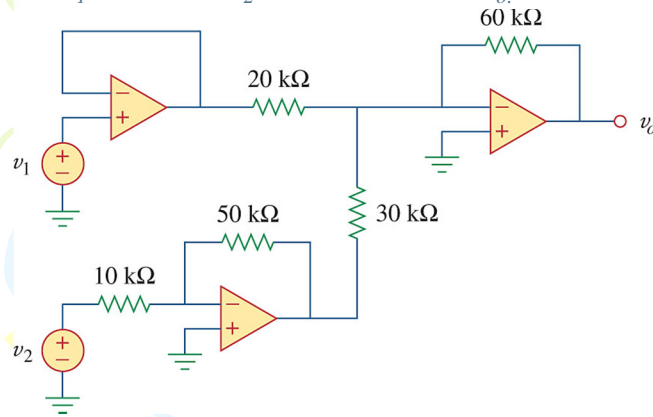
29

PP 5-10If $v_1 = 4 \text{ V}$ and $v_2 = 3 \text{ V}$, determine v_o .

30

PP 5-10 (continued)

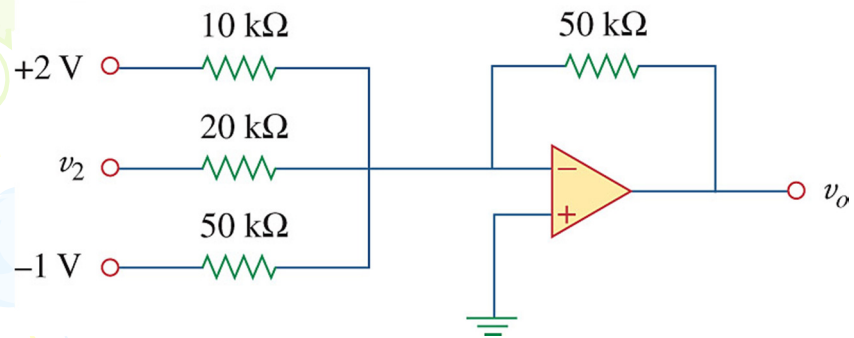
If $v_1 = 4\text{ V}$ and $v_2 = 3\text{ V}$, determine v_o .



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Problem 5-39

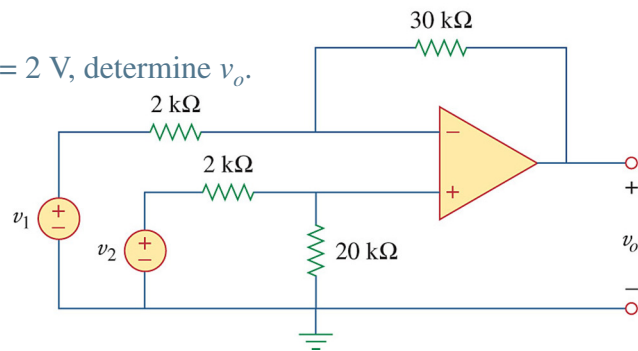
Determine v_2 such that $v_o = -16.5\text{ V}$.



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Problem 5-39

If $v_1 = 1$ V and $v_2 = 2$ V, determine v_o .

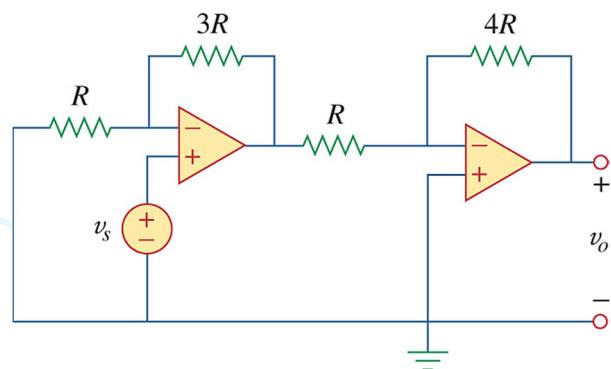


$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

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Problem 5-59

Determine v_o/v_s if $R = 20$ kΩ.



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